QForge System Demonstration & Operating Playbook

Comprehensive, Step-by-Step Viva & Faculty Presentation Manual for `app_web.py` & `app.py`

Project Title	Hybrid Quantum-Classical-Quantum CNN for Brain MRI (QCQ-CNN)	**Target Deliverables**	Clinical Web Studio & Desktop Lab Console
Team Lead & Presenter	Yashwanth S (2024BCS0264)	**Project Team**	Agnivesh Anil Warrier, N. Vishnuvardhan Reddy, B. Bharath Kumar, A. Krishna Bharadwaj Reddy
Benchmark Platform	Deduplicated Kaggle Brain MRI (5,222 Clean Slices)	**Verification Level**	23/23 Pytest Unit Tests Passing (Exact Parity with PennyLane)

1.0 QUICK-START LAUNCHER CHEAT-SHEET

🌐 1. Interactive Web Studio (app_web.py)

Best for: **Examiner walkthroughs, 1-click clinical inference, quantum feature maps, live NISQ depolarizing noise tests.**

```
python app_web.py
```

Auto-opens in browser at: `http://localhost:8501`
Alternate launcher: Double-click `run_web.bat`

🖥️ 2. Desktop Lab Workstation (app.py)

Best for: **Training automation, live loss chart dynamics, 434 img/s throughput telemetry, statistical p-value matrices.**

```
python app.py
```

Launches dark-mode Qt GUI (AMD Ryzen / Intel optimized)
Alternate launcher: Double-click `run_app.bat`

2.0 RECOMMENDED 5-MINUTE LIVE DEMO AGENDA

Time	Phase	Action Performed on Screen	Examiner Takeaway / Value
0:00 - 0:45	Architecture Context	Explain QCQ-CNN pipeline: 4Q Filter → Classical CNN → 2Q VQC Head.	Clear problem framing: resolving Nature 2025 single-qubit bottleneck.
0:45 - 2:30	**Web Studio Demo** (`app_web.py`)	1. Click `[ LOAD: GLIOMA SAMPLE ]` → Malignant outcome. 2. Explain 4-channel quantum feature maps. 3. Click `[ LOAD: MENINGIOMA SAMPLE ]` → Benign. 4. Toggle **Simulated Noise** (γ=0.02) to prove Model B resilience.	Interactive clinical diagnosis, real-time quantum kernel inspection, fault tolerance under hardware noise.
2:30 - 4:00	**Desktop Lab Demo** (`app.py`)	1. Navigate to **Tab 2 (RUNNER)**. 2. Hit `[ > RUN EXPERIMENT ]` (smoke test) → show 434 img/s. 3. Switch to **Tab 3 (RESULTS)** → show 5-seed p=0.220 table.	Rigorous engineering proof: 434 img/s speedup via pure-PyTorch einsum and statistical rigor.
4:00 - 5:00	Viva & Q&A Defense	Answer faculty questions using Chapter 5 mathematical proofs and deduplication audit logs.	Demonstrates deep mastery of both quantum circuits and ML data integrity.

Pre-Demo Verification: Both tools run locally on CPU without cloud credentials. The virtual environment includes PyTorch, PennyLane, Streamlit, and PySide6.

PART I: CLINICAL WEB STUDIO WALKTHROUGH

Step-by-step Operator Script for `app_web.py`

INTERFACE 1 (WEB)

3.0 INTERACTIVE WEB STUDIO WALKTHROUGH (APP_WEB.PY)

The Web Studio provides an intuitive, high-impact clinical instrument panel configured for live demonstration.

STEP 1: INITIATE WEB APPLICATION

TERMINAL COMMAND

```
python app_web.py
```

The system detects execution mode, binds to `http://localhost:8501`, and automatically opens your web browser.

STEP 2: RUN CLINICAL INFERENCE (GLIOMA NEOPLASM EXAMPLE)

TAB 1: LIVE INFERENCE

1. On **Tab 1 (`> 01 // LIVE INFERENCE & KERNELS`)**, click the button: `[ LOAD: GLIOMA SAMPLE ]`.
2. Observe the 3 synchronized visual panels rendered instantly on screen:

- **Panel 1 (Left — 28×28 Normalized Input):** Normalized MRI slice showing the hyperintense left-hemisphere infiltrating glioma lesion.
- **Panel 2 (Center — 4 Quantum Filter Channels):** Displays the 14×14 Pauli expectations [⟨Z0⟩, ⟨Z1⟩, ⟨Z2⟩, ⟨Z3⟩]. Point out how Wire 0 captures luminance topology while Wires 1–3 isolate tumor boundary gradients.
- **Panel 3 (Right — Diagnostic Outcome Card):** Highlights the red `MALIGNANT NEOPLASM` badge with high diagnostic confidence.

> **What to Say to Evaluators:**
> *"Here we pass an MRI slice through our 4-qubit quantum convolutional filter. Each 2×2 pixel patch is angle-encoded into four qubits, transformed via trainable parameterized unitary gates, and measured in the Pauli-Z basis. This yields 4 orthogonal quantum feature channels that feed our classical backbone."*

STEP 3: COMPARE WITH BENIGN SAMPLE & INSPECT STATE

QUANTUM INTROSPECTION

1. Click `[ LOAD: MENINGIOMA SAMPLE ]`.
2. Watch the diagnostic card transition to green `BENIGN NEOPLASM`.
3. Scroll down to **"Terminal VQC Quantum State Introspection"**:

- Observe the latent angles: h0, h1 in [-π, π] produced by the classical CNN backbone.
- Observe the measured observables: Single-qubit ⟨Z0⟩, ⟨Z1⟩, and the joint entanglement observable **⟨Z0 Z1⟩**.

> **What to Say to Evaluators:**
> *"Notice the lower introspection panel. In previous literature, the second qubit and the joint correlation ⟨Z0 Z1⟩ were completely discarded by measuring only ⟨Z0⟩. Our architecture explicitly extracts ⟨Z0 Z1⟩, directly preserving two-qubit entanglement information for the final diagnostic decision."*

STEP 4: LIVE NISQ HARDWARE NOISE STRESS-TEST

SIDEBAR CONTROLS

1. In the left sidebar, check **"Enable Simulated Quantum Noise"**.
2. Set **Depolarizing Rate γ** to `0.02` and **Measurement Shots** to `1024`.
3. Switch between **Model A (Frozen Random)** and **Model B + Correlated Readout**:

- **Model A:** Accuracy deteriorates dramatically under noise (experiences a -10.54% drop).
- **Model B + Correlated:** Remains resilient with only a -2.75% deviation, maintaining reliable diagnostic output.

> **What to Say to Evaluators:**
> *"To prove viability on noisy intermediate-scale quantum (NISQ) hardware, we simulate depolarizing gate noise and finite measurement shot noise. Notice that our correlated architecture retains over 84.5% diagnostic accuracy even under typical NISQ hardware noise levels."*

PART II: DESKTOP RESEARCH WORKSTATION WALKTHROUGH

Step-by-step Operator Script for `app.py`

INTERFACE 2 (DESKTOP)

4.0 DESKTOP RESEARCH WORKSTATION WALKTHROUGH (APP.PY)

The PySide6/Qt desktop instrument panel is engineered for reproducible laboratory experiment dispatch, telemetry logging, and statistical matrix verification.

STEP 1: INITIATE DESKTOP WORKSTATION

TERMINAL COMMAND

```
python app.py
```

Launches the high-performance Quantum-Lab research interface styled with dark theme (#060910 canvas) and monospace typography.

STEP 2: TAB 1 (DASHBOARD) — REVIEW EXPERIMENTAL AUDIT

DASHBOARD NAVIGATION

Direct the evaluators' attention to the 4 instrument stat cards across the top:

- `MODEL A · FROZEN: 85.75%` — Baseline performance with frozen random unitary reservoir.
- `MODEL B · TRAINABLE: 85.36%` — Differentiable filter with identical single-qubit measurement.
- `PARAMETER DRIFT: 1.83 RAD` — Direct mathematical proof of non-vanishing gradient descent.
- `VERDICT: p = 0.220` — Confirms no statistically significant difference on single-qubit readout.

> **What to Say to Evaluators:**
> *"Our dashboard summarizes 25 rigorous seed runs on the leak-free dataset. Notice the p-value of 0.220: when using single-qubit readout, a frozen random filter achieves identical classification to a trained filter, proving the Quantum Reservoir Principle. The bottleneck was never the filter; it was the readout."*

STEP 3: TAB 2 (RUNNER) — DISPATCH REAL-TIME EXPERIMENT

LIVE TELEMETRY MONITORING

1. Switch to **Tab 2 (RUNNER)**.
2. Set `Filter Mode: trainable`, `Seed: 42`, `Max Epochs: 2`.
3. Click the cyan button: `[ > RUN EXPERIMENT ]`.
4. Point out the live telemetry features:

- **Execution Console:** Tagged log streams with `[SYS]`, `[TRAIN]`, `[DATA]`, `[OK]` indicators and live timestamping.
- **Throughput:** Highlighting our **434 img/s pure-PyTorch vectorized engine**, completing an entire epoch in under 7 seconds on CPU.
- **Live Dynamics Line Chart:** Training loss (cyan dashed) and validation loss (violet solid) render dynamically per epoch.

> **What to Say to Evaluators:**
> *"Observe the training speed: by vectorizing PennyLane statevector math into pure PyTorch einsum operations, we achieved 434 images per second on standard CPU. Training an epoch now takes 7 seconds instead of 30+ minutes, enabling full 5-seed statistical verification."*

STEP 4: TAB 3 & TAB 4 — STATISTICAL VERIFICATION & FIGURES

RESULTS & ARTIFACTS

- **Tab 3 (RESULTS):** Click `[ SIGNIFICANCE ]` to display paired t-test and Wilcoxon signed-rank tables.
- **Tab 4 (VISUALIZATIONS):** Click through publication figures:
  - `Fig 1 (Learning Dynamics)`: Loss curves over 15 epochs.
  - `Fig 2 (Parameter Drift)`: Shows steady weight evolution away from θ0.
  - `Fig 3 (ROC & Confusion)`: 88.90% AUC achieved by Correlated architecture.
  - `Fig 6 (Noise Resilience)`: Accuracy vs depolarizing rate γ in [0.00, 0.05].

5.0 EXAMINER TECHNICAL DEFENSE QUESTIONS & ANSWERS

Q1: Why does QForge report ~87.25% test accuracy when Long et al. (Nature Sci Rep 2025) reported 95.18%?

Definitive Defense: *"The published Nature 2025 study evaluated on raw Kaggle partitions that contained **110 cross-split duplicates** between the training and test sets. This introduced severe data leakage, causing artificial score inflation. We conducted a rigorous perceptual hash audit (`audit_and_deduplicate()`) using dhash (threshold ≤ 6). On the leak-free, clean dataset, the true baseline accuracy is 85.75%, which our correlated readout model elevates to 87.25% (and AUC to 88.90%)."*

Q2: Why did making the front quantum filter trainable not beat the frozen filter on single-qubit readout?

Definitive Defense: *"This was one of our primary research findings: **The Quantum Reservoir Principle**. Because the input patches are 2×2 pixels, a frozen random unitary circuit acts as an optimal, zero-cost non-linear kernel projector into Hilbert space. When the terminal readout only measures single observable ⟨Z0⟩, the model's expressivity is strictly bottlenecked at the output. Gradient descent shifts the filter weights (proven by parameter drift $||\theta - \theta 0|| \approx 1.83$ rad), but the single-qubit measurement cannot capture the extra expressive capacity (p = 0.220)."*

Q3: How does your Correlated Readout solve the information bottleneck?

Definitive Defense: *"The terminal 2-qubit VQC uses an entangling CNOT gate to synthesize two-qubit quantum correlations. However, standard architectures trace out qubit 1 and only measure ⟨Z0⟩. Our correlated head measures the full observable basis: [⟨Z0⟩, ⟨Z1⟩, ⟨Z0 Z1⟩]. The joint expectation ⟨Z0 Z1⟩ = (P00 + P11) - (P01 + P10) preserves non-local entanglement, improving test accuracy to 87.25% and AUC to 88.90%."*

Q4: Does your model suffer from Barren Plateaus?

Definitive Defense: *"No. Barren plateaus occur primarily in deep circuits with random Haar unitaries across high qubit counts. Our architecture uses shallow circuits (L=1 to 3 layers) over 4 qubits in the filter and 2 qubits in the head, with localized entanglement. Parameter drift metrics confirm active, healthy gradient magnitudes throughout training."*

6.0 LIVE DEMO EMERGENCY TROUBLESHOOTING

Potential Issue	Root Cause	Immediate 5-Second Fix
Port 8501 Busy	Another Streamlit instance is running in background.	Run: `python app_web.py --server.port 8502`
Virtual Environment Not Active	Using system Python instead of project packages.	Run: `.\.venv\Scripts\Activate.ps1`
Verification Check Required	Examiner asks to prove code validity on the spot.	Run: `pytest tests/test_qsim_and_models.py -v` (14 tests pass in 3s)

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